Exelon Generation.

November 17, 2017

10 CFR 50.73

SVP-17-069

U. S. Nuclear Regulatory Commission
ATTN: Document Control Desk
Washington, D.C. 20555

Quad Cities Nuclear Power Station, Units 1 and 2
Renewed Facility Operating License Nos. DPR-29 and DPR-30
NRC Docket Nos. 50-254 and 50-265

Subject: Licensee Event Report 254/2017-003-00, "Control Room Emergency Ventilation Air Conditioning Piping Refrigerant Leak Due to High Cycle Fatigue"

Enclosed is Licensee Event Report (LER) 254/2017-003-00, "Control Room Emergency Ventilation Air Conditioning Piping Refrigerant Leak Due to High Cycle Fatigue," for Quad Cities Nuclear Power Station, Unit 1.

This report is submitted in accordance with 10 CFR 50.73 (a)(2)(v)(D) which requires the reporting of any operation or condition prohibited by Technical Specifications.

There are no regulatory commitments contained in this letter.

Should you have any questions concerning this report, please contact Rachel A Luebbe at (309) 227-2813.

Respectfully,

Kenneth Ohr
Site Vice President
Quad Cities Nuclear Power Station

cc: Regional Administrator – NRC Region III
NRC Senior Resident Inspector – Quad Cities Nuclear Power Station

NRC FORM 366 (04-2017) U.S. NUCLEAR REGULATORY COMMISSION

LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/)

1. FACILITY NAME	2. DOCKET NUMBER	3. PAGE
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4. TITLE
Control Room Emergency Ventilation Air Conditioning Piping Refrigerant Leak Due to High Cycle Fatigue

5. EVENT DATE			6. LER NUMBER			7. REPORT DATE			8. OTHER FACILITIES INVOLVED	
MONTH	DAY	YEAR	YEAR	SEQUENTIAL NUMBER	REV NO.	MONTH	DAY	YEAR	FACILITY NAME: Quad Cities Unit 2	DOCKET NUMBER: 05000265
09	21	2017	2017	- 003 -	00	11	17	2017	FACILITY NAME: N/A	DOCKET NUMBER: N/A

9. OPERATING MODE: 1

10. POWER LEVEL: 100

11. THIS REPORT IS SUBMITTED PURSUANT TO THE REQUIREMENTS OF 10 CFR §: *(Check all that apply)*

☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)
☐ 20.2203(a)(2)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)
☐ 20.2203(a)(2)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(v)(A)	☐ 73.71(a)(4)
☐ 20.2203(a)(2)(iii)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(v)(B)	☐ 73.71(a)(5)
☐ 20.2203(a)(2)(iv)	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(1)
☐ 20.2203(a)(2)(v)	☐ 50.73(a)(2)(i)(A)	☒ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(i)
☐ 20.2203(a)(2)(vi)	☐ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(vii)	☐ 73.77(a)(2)(ii)
	☐ 50.73(a)(2)(i)(C)	☐ OTHER Specify in Abstract below or in NRC Form 366A	

12. LICENSEE CONTACT FOR THIS LER

LICENSEE CONTACT	TELEPHONE NUMBER *(Include Area Code)*
Rachel Luebbe – Regulatory Assurance	(309)-227-2813

13. COMPLETE ONE LINE FOR EACH COMPONENT FAILURE DESCRIBED IN THIS REPORT

CAUSE	SYSTEM	COMPONENT	MANU-FACTURER	REPORTABLE TO EPIX	CAUSE	SYSTEM	COMPONENT	MANU-FACTURER	REPORTABLE TO EPIX
X	VI	PSF	N/A	Y					

14. SUPPLEMENTAL REPORT EXPECTED	15. EXPECTED SUBMISSION DATE	MONTH	DAY	YEAR
☐ YES *(If yes, complete 15. EXPECTED SUBMISSION DATE)* ☒ NO		N/A	N/A	N/A

ABSTRACT *(Limit to 1400 spaces, i.e., approximately 15 single-spaced typewritten lines)*

On 09/21/2017 at 1550, Operations started Control Room Emergency Ventilation (CREV) Air Conditioning (AC) for Tracer Gas Testing. Per Operations instructions, Mechanical Maintenance went to inspect the Control Room Emergency Ventilation system for refrigerant leaks before Tracer Gas Testing was started. Mechanical Maintenance reported a refrigerant leak on the discharge piping of the compressor, right above the inlet to the condenser. The leak was at the expansion joint of a fitting. The safety significance of this event was minimal.

The cause of the refrigerant leak on the Control Room Emergency Ventilation compressor discharge pipe fitting into the condenser was due to high cycle fatigue.

The corrective action was to replace the failed Control Room Emergency Ventilation compressor discharge pipe fitting.

The CREV AC system is a single train system. Given the impact on the CREV AC system, this report is submitted in accordance with the requirements of 10 CFR 50.73(a)(2)(v)(D), which requires the reporting of any event or condition that could have prevented the fulfillment of the safety function of structures or systems that are needed to mitigate the consequences of an accident.

NARRATIVE

PLANT AND SYSTEM IDENTIFICATION

General Electric - Boiling Water Reactor, 2957 Megawatts Thermal Rated Core Power

Energy Industry Identification System (EIIS) codes are identified in the text as [XX].

EVENT IDENTIFICATION

Control Room Emergency Ventilation Air Conditioning Compressor Discharge Pipe Fitting Refrigerant Leak

A. CONDITION PRIOR TO EVENT

Unit: 1	Event Date: September 21, 2017	Event Time: 1550 hours
Reactor Mode: 1	Mode Name: Power Operation	Power Level: 100%

B. DESCRIPTION OF EVENT

On 09/21/2017 at 1550, Operations started Control Room Emergency Ventilation (CREV) Air Conditioning (AC)[CMP] for Tracer Gas Testing. Per Operations instructions, Mechanical Maintenance went to inspect the Control Room Emergency Ventilation system for refrigerant leaks before Tracer Gas Testing was started. Mechanical Maintenance reported a refrigerant leak on the discharge piping [PSF] of the compressor, right above the inlet to the condenser [COND]. The leak was at the expansion joint of a fitting. Operations declared the refrigeration condensing unit (RCU) of Train B of Control Room HVAC inoperable at 1730 and entered Technical Specification (TS) 3.7.5 Condition A. The refrigerant leak was minimal, which allowed the Air Conditioning Compressor to run and maintain Control Room Temperature before it was secured. This event is a Maintenance Rule Functional Failure for the Control Room Emergency Ventilation system.

C. CAUSE OF EVENT

The cause of the refrigerant leak on the Control Room Emergency Ventilation Air Conditioning compressor discharge pipe fitting was due to a high cycle fatigue crack that initiated at the root of expansion joint transition on the outer surface. No material anomalies were associated with the crack origin. The compressor discharge piping is approximately five linear feet in length with two 90-degree bends and enters the condenser at a 45-degree angle. Adding a support to the short length of pipe would not reduce the vibrations or stress acting on the pipe. A contributing factor to the high cycle fatigue was that the lifespan (20 yrs) of the fitting was reduced because of high vibration conditions it experienced during a portion of its installed life. Three previous compressor failures (2005, 2009 & 2011) caused the high vibration conditions.

D. SAFETY ANALYSIS

System Design

The purpose of the Control Room Emergency Ventilation system is to maintain the proper air environment for instrumentation and personnel in the Control Room. The Train B CREV AC system is a single train safety-related system that is designed to maintain design temperature in the Control Room Envelope (CRE) under post-accident

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conditions. The loss of the CREV AC could impact the plant's ability to mitigate the consequences of an accident. The Train B CREV Refrigeration Condensing Unit (RCU) 0-9400-102 is a Carrier reciprocating compressor; Model 5H120, with a capacity of 90 tons. The backup to the Train B CREV system is the Train A Control Room HVAC system, which is a non-safety-related system.

Safety Impact

The safety significance of this event was minimal. The CREV system filtration capability was not lost, and control room temperature was maintained during this event. In addition, the non-safety related Train A Control Room HVAC system was available throughout this event. Although the refrigerant leakage was small, CREV AC could not be relied upon to fully perform its safety function during an accident.

Risk Insights

The CREV AC system is not explicitly modeled in the Plant Probabilistic Risk Assessment (PRA). Certain Operator Actions (OAs) are modeled in the PRA; however, the effect of this event on OA human error probabilities (HEPs) would have a negligible quantitative impact on the calculated Core Damage Frequency (CDF) and Large Early Release Frequency (LERF) since loss of CREV AC events reduce temperature control of the Control Room temperature as slow progressing events. In the loss of CREV system event, station procedures direct the opening of doors and use of temporary fans, as necessary. Applicable operator training is performed on these actions, and the equipment is pre-staged as a part of the station blackout response requirements.

Although opening Control Room doors to support room cooling would introduce the increased potential for radiological or chemical events impacting the Main Control Room habitability, the probability of a loss of Control Room HVAC event coincident with a radiological or chemical event is small and therefore, would also have a negligible quantitative impact on CDF and LERF.

In conclusion, the degradation of the 0-9400-102 CREV AC compressor discharge pipe fitting was not risk significant.

E. CORRECTIVE ACTIONS

1. The failed Control Room Emergency Ventilation compressor discharge pipe fitting was replaced.

2. Since the three previous compressor failures (2005,2009 & 2011) that caused high vibration conditions were corrected via a compressor modification, the new compressor discharge pipe fitting is expected to function beyond the life of the plant.

F. PREVIOUS OCCURRENCES

The station events database, LERs, and INPO ICES were reviewed for similar events at Quad Cities Nuclear Power Station. This event was primarily attributed to high cycle fatigue on the Control Room Emergency Ventilation Air Conditioning compressor discharge pipe fitting.

No previous occurrences applicable to the circumstances of this event were identified in this search.

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G. COMPONENT FAILURE DATA

Failed Equipment: Control Room HVAC Compressor Discharge/Condenser Inlet Pipe Fitting
Component Manufacturer: Mueller Brass Company
Component Model Number: W-1381

This event has been reported to ICES under record #423784.